

WEBSITE TRAFFIC ANALYSIS REPORT

Data Analytics Project Report

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Role: Data Analyst

Project Type: Exploratory Data Analysis (EDA) — Website Traffic & Conversion Performance

2. Executive Summary

This report presents a comprehensive analysis of website traffic performance covering six months of data (February through July 2026), totaling 9,003 traffic records across five acquisition channels and ten site pages, representing 109,463 sessions and 98,094 users. The analysis was conducted to identify which traffic sources and pages drive the strongest engagement and conversion outcomes, with the objective of informing marketing and content investment decisions. Following a structured data-cleaning process to resolve missing values, duplicate records, invalid metric values, and inconsistent date formats, the analysis found an overall conversion rate of 4.20% and an average bounce rate of 47.2%, with Paid Search delivering the highest conversion rate (5.59%) despite Organic Search generating the largest volume of traffic (35.5% of all sessions). Bounce rate showed a clear negative relationship with conversions, and the Checkout and Pricing pages emerged as the strongest converters, while blog and informational pages showed the highest bounce rates and lowest conversion rates. These findings support a targeted strategy: reallocating incremental spend toward higher-converting channels, reducing bounce on top-of-funnel content pages, and strengthening the path from informational content to high-converting product and checkout pages.

3. Introduction

3.1 Background

Digital marketing and product teams increasingly rely on traffic and engagement data to understand which channels and pages generate genuine business value, rather than simply the highest volume. Raw session counts alone do not indicate whether traffic is converting or where visitors are disengaging. This project was undertaken to convert raw website analytics into a structured, evidence-based understanding of traffic quality, engagement, and conversion performance.

3.2 Objective

The primary objective of this analysis is to evaluate the performance of each traffic source and page, identify the factors most closely associated with conversions and bounce rate, and surface actionable, specific recommendations that marketing and web teams can implement to improve overall site performance.

3.3 Scope of Analysis

The scope of this analysis covers six months of traffic data (February 1, 2026 through July 31, 2026) across five acquisition channels — Organic Search, Paid Search, Referral, Social, and Direct — and ten site pages spanning the homepage, product pages, blog content, informational pages, pricing, and checkout. The

analysis includes data cleaning, descriptive statistics, trend analysis, source and page-level comparison, correlation analysis, and identification of underperforming pages. The report does not extend to attribution modeling across multiple touchpoints or paid media cost data — findings are based on session-level engagement and conversion metrics as provided.

4. Data Overview

The dataset used for this analysis is structured at the date-source-page level, meaning each row represents one traffic source's activity on one page for one day. This structure enables granular, page- and channel-specific analysis while still allowing aggregation to daily, monthly, and channel-wide totals.

4.1 Dataset Summary

Metric	Value
Total records (date-source-page rows)	9,003
Time period covered	February 1, 2026 - July 31, 2026 (6 months)
Traffic sources covered	5 (Organic Search, Paid Search, Referral, Social, Direct)
Pages covered	10 (Home, product, blog, informational, pricing, checkout)
Total sessions	109,463
Total users	98,094

4.2 Key Fields

Field	Description
date	Date of the traffic record
traffic_source	Organic Search / Paid Search / Referral / Social / Direct
page_url	Page path the traffic landed on
sessions	Number of sessions
users	Number of unique users
bounce_rate	Share of single-page sessions
session_duration	Average session duration (seconds)
pages_per_session	Average pages viewed per session
conversions	Number of conversion events
goal_completions	Number of goal completions (incl. micro-conversions)

5. Methodology

5.1 Data Cleaning

Prior to analysis, the raw dataset was systematically cleaned to ensure accuracy and reliability of findings:

- **Missing values:** Missing bounce rate, session duration, pages-per-session, and conversion values were identified and imputed using traffic-source-and-page-level averages, with an overall fallback average where group-level data was unavailable.
- **Duplicate records:** Exact duplicate rows were identified and removed.
- **Date parsing:** The date field contained multiple inconsistent formats (e.g., DD/MM/YYYY, MM-DD-YYYY, and written-out dates); all values were parsed into a single consistent date type.
- **Invalid values:** Negative session counts and session durations were treated as tracking errors and removed; bounce rate values outside the valid 0-100% range were capped to valid bounds.
- **Data type consistency:** Fields such as pages_per_session, which contained mixed numeric/text entries, were standardized to a consistent numeric type.
- **Text standardization:** The traffic_source field was trimmed and normalized for consistent casing prior to grouping and aggregation.

5.2 Analytical Approach

The analysis followed a structured exploratory data analysis (EDA) workflow: sessions were aggregated by day and month to identify trends; performance was compared across traffic sources and pages; correlation coefficients were calculated to quantify the relationship between bounce rate, session duration, pages-per-session, and conversions; sessions were segmented into low- and high-bounce-rate groups to compare conversion outcomes; and month-over-month growth rates were calculated to track traffic trajectory.

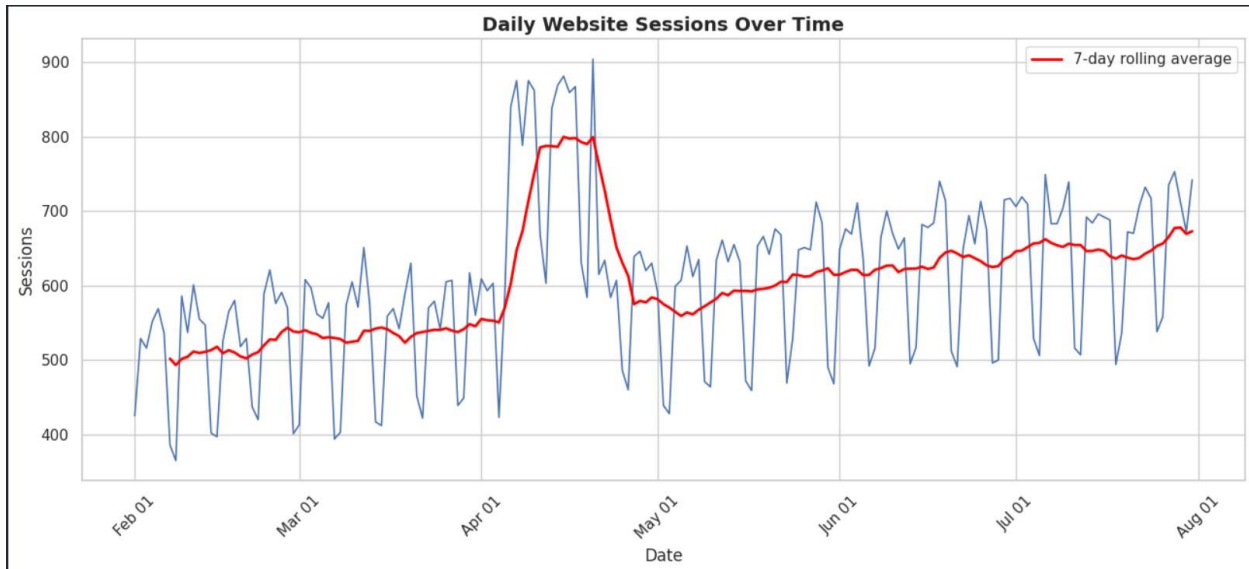
5.3 Tools Used

Tool	Purpose
Python (pandas, numpy)	Data cleaning, transformation, and statistical computation
Matplotlib & Seaborn	Data visualization (charts, plots, heatmaps)
Jupyter Notebook	Analysis environment and documentation
SQL	Structured querying, trend analysis, and grouped aggregation

6. Key Findings

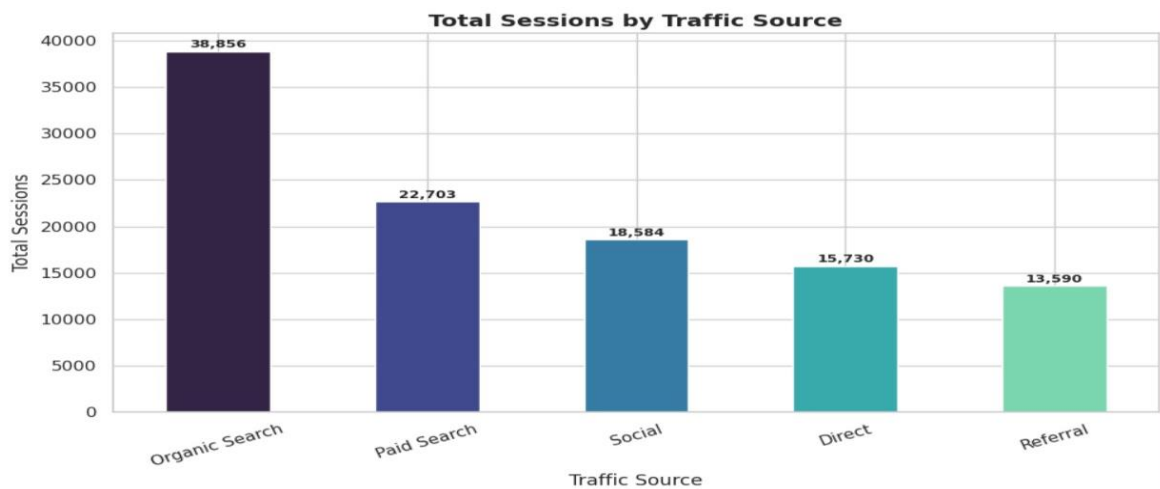
6.1 Overall Traffic Trends (Daily / Monthly Patterns)

Monthly sessions grew from 14,425 in February to 20,440 in July, though the trajectory was not strictly linear. April recorded the highest monthly total (20,682 sessions, +24.3% month-over-month), driven by a marketing campaign concentrated in Paid Search and Social channels. May saw a notable pullback (-11.8% month-over-month) following the campaign's conclusion, before traffic resumed a steady upward trend through June and July. This pattern indicates that campaign-driven spikes are temporary unless reinforced by sustained organic or owned-channel growth.



6.2 Traffic Source Performance Comparison

Organic Search is the largest single channel, contributing 35.5% of total sessions (38,856 sessions), followed by Paid Search (20.7%, 22,703 sessions), Social (17.0%, 18,584 sessions), Direct (14.4%, 15,730 sessions), and Referral (12.4%, 13,590 sessions). However, session volume and conversion efficiency diverge sharply: Paid Search converts at 5.59%, the highest of all channels, while Social converts at just 2.10%, the lowest — despite Social outperforming Referral and Direct on raw session volume.



6.3 User Engagement Analysis

Engagement metrics vary considerably by channel. Direct traffic shows the strongest engagement, with an average session duration of 233 seconds and 2.80 pages per session, followed closely by Referral (213 seconds, 2.60 pages/session) and Organic Search (199 seconds, 2.46 pages/session). Paid Search and Social show comparatively weaker engagement — Social in particular averages just 108 seconds and 1.79 pages per session, the lowest of any channel, reinforcing that Social visitors are less likely to explore the site in depth.

6.4 Conversion Rate Analysis by Traffic Source and by Page

At the channel level, conversion rate ranks Paid Search (5.59%) highest, followed by Direct (4.68%), Organic Search (4.25%), Referral (4.04%), and Social (2.10%) lowest. At the page level, the /checkout page converts at 9.74% — nearly double the next-best page — followed by /pricing (5.90%) and the three product pages (running shoes 5.48%, winter jackets 5.09%, backpacks 5.02%). The homepage converts at a moderate 3.95%, while blog and informational pages (/blog/how-to-choose-running-shoes at 2.64%, /blog/summer-sale-guide at 2.10%, /about-us at 1.98%, and /contact at 1.85%) convert well below the site average of 4.20%.

6.5 Top 5 High-Performing and Bottom 5 Low-Performing Pages

Ranked by conversion rate, the following pages represent the strongest and weakest performers on the site:

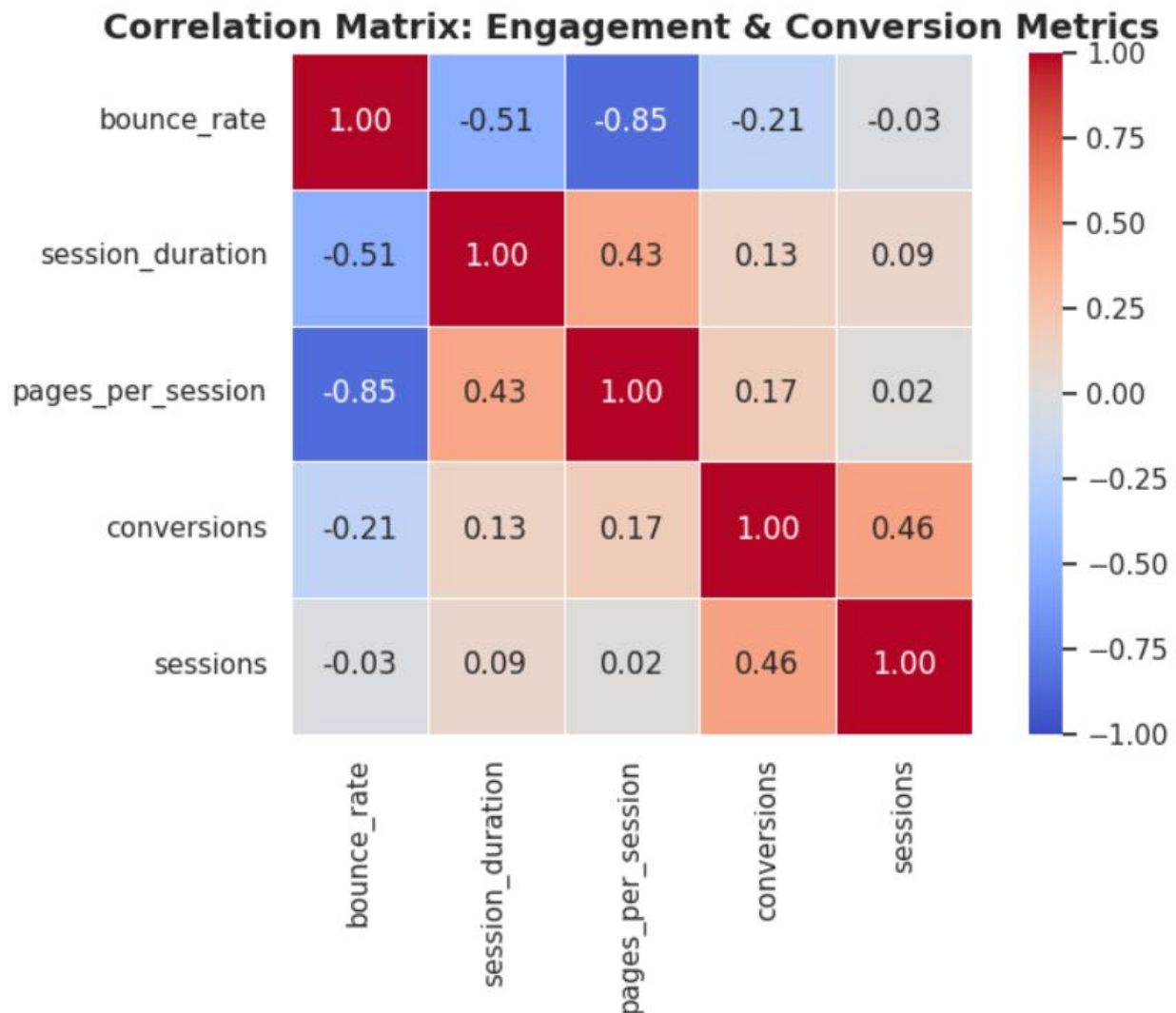
Rank	Page	Sessions	Conversion Rate
1 (Top)	/checkout	4,536	9.74%
2	/pricing	7,676	5.90%
3	/product/running-shoes	16,793	5.48%
4	/product/winter-jackets	12,268	5.09%
5	/product/backpacks	8,857	5.02%

Rank	Page	Sessions	Conversion Rate
1 (Bottom)	/contact	5,447	1.85%
2	/about-us	6,712	1.98%
3	/blog/summer-sale-guide	11,270	2.10%
4	/blog/how-to-choose-running-shoes	13,576	2.64%
5	/home	22,328	3.95%

Notably, the homepage receives the highest session volume of any page (22,328 sessions) but converts below the site average, making it a high-leverage candidate for on-page optimization given its reach.

6.6 Key Factors Influencing Bounce Rate and Conversions

Correlation analysis shows a clear negative relationship between bounce rate and conversions ($r = -0.21$): sessions in the lower half of the bounce-rate distribution convert at 5.41%, nearly double the 2.83% conversion rate observed in the higher-bounce-rate half. Pages-per-session shows a positive relationship with conversions ($r = 0.17$), confirming that deeper on-site engagement is associated with a higher likelihood of conversion. While these correlations are moderate rather than extremely strong, they are directionally consistent across every cut of the data — by page, by channel, and by month — supporting their reliability as decision-making signals.



7. Insights & Business Implications

Insight 1 — Organic Search leads on volume but not on efficiency. Organic Search delivers the largest share of sessions (35.5%) but converts below Paid Search and Direct. For marketing leadership, this means organic strength should be treated primarily as a brand-visibility and top-of-funnel asset, with conversion optimization efforts focused elsewhere in the funnel rather than solely on driving more organic volume.

Insight 2 — Paid Search is the most efficient channel and merits continued investment. With the highest conversion rate (5.59%) despite moderate volume, Paid Search demonstrates the strongest return per session of any channel. This supports maintaining or cautiously increasing paid budget, provided cost-per-acquisition remains within target thresholds — a factor outside the scope of this session-level dataset.

Insight 3 — Bounce rate is a leading indicator of conversion performance, not just an engagement metric. The near two-times gap in conversion rate between low- and high-bounce-rate sessions confirms that bounce rate reduction has a direct, measurable link to revenue-driving outcomes — not merely a proxy for page quality.

Insight 4 — Social traffic underperforms on both engagement and conversion despite solid volume. Social sessions show the shortest average duration, fewest pages per session, and lowest conversion rate of any channel. For decision-makers, this signals a mismatch between the audience Social channels attract and the on-site experience — worth addressing through either landing-page alignment or a shift toward brand-awareness goals rather than direct-response goals.

Insight 5 — The homepage is a high-reach, under-converting asset. As the single highest-traffic page on the site, even a modest improvement in the homepage's below-average conversion rate would yield a disproportionately large impact on total site conversions, given its reach.

Insight 6 — Blog and informational content attracts traffic but does not currently guide it toward conversion. Blog posts and the About/Contact pages show the highest bounce rates and lowest conversion rates on the site. This is expected for top-of-funnel content, but represents an untapped opportunity: this traffic is not currently being routed toward higher-converting product or pricing pages.

Insight 7 — Checkout and Pricing pages validate the bottom-of-funnel experience. These two pages post the highest conversion rates on the site by a wide margin, confirming the purchase-intent experience is functioning well once visitors reach it. The primary opportunity lies in getting more qualified visitors to these pages, not in redesigning them.

8. Recommendations

8.1 Reallocate Marketing Investment Toward Higher-Converting Channels

- Incrementally shift a portion of Social budget toward Paid Search, subject to cost-per-acquisition analysis, given the significant conversion-rate gap between the two channels.

- Treat Organic Search as a volume and brand-visibility channel, with SEO investment paired with on-page conversion optimization rather than traffic growth alone.

8.2 Reduce Bounce Rate on High-Traffic, Low-Conversion Pages

- Prioritize UX and content improvements on the homepage first, given its combination of high traffic and below-average conversion rate.
- Add clear, prominent calls-to-action and simplified navigation on blog and informational pages to reduce single-page exits.

8.3 Strengthen the Path from Content to Conversion

- Add contextual internal links from blog posts to relevant product pages (e.g., linking the running-shoes blog post directly to the running-shoes product page).
- Introduce mid-content or exit-intent prompts directing About/Contact page visitors toward Pricing or relevant product pages.

8.4 Re-Evaluate the Social Channel Strategy

- Audit landing pages used for Social campaigns to ensure message-to-page alignment with what the ad or post promised.
- Consider adjusting Social channel KPIs to reflect awareness and reach rather than direct conversion, if on-site behavior does not improve after landing-page testing.

8.5 Monitor Trend and Seasonality, Not Just Channel Totals

- Track month-over-month growth rates alongside channel totals to distinguish sustained growth from temporary campaign-driven spikes, as seen in the April surge and May pullback.
- Set a cadence (e.g., monthly) for reviewing whether post-campaign traffic settles at a new, higher baseline or reverts to pre-campaign levels.

8.6 Use Engagement Metrics as Early Warning Indicators

- Track pages-per-session and bounce rate alongside conversion rate in reporting dashboards, since both correlate meaningfully with conversion outcomes and can surface issues before conversion data fully reflects them.

9. Conclusion

This analysis of six months of website traffic across five acquisition channels and ten pages confirms that traffic volume and traffic quality are distinct dimensions of performance — the channels and pages that draw the most visitors are not always the ones that convert them most effectively. Paid Search and the Checkout/Pricing pages stand out as the most efficient parts of the funnel, while Social traffic and

blog/informational content represent the clearest opportunities for improvement. The consistent, moderate-to-strong relationship between bounce rate, on-site engagement, and conversions across every cut of the data provides a reliable evidence base for the recommendations outlined in this report.

The recommendations are intended to be implemented together: channel reallocation addresses acquisition efficiency directly, while bounce-rate reduction and stronger internal linking address the on-site experience that determines whether acquired traffic ultimately converts. As a next step, it is recommended that the marketing and web teams pilot the proposed homepage and blog-linking improvements for one quarter, then re-run this analysis to measure the resulting impact on bounce rate and conversion rate. Extending the dataset in future periods to include device type, geography, campaign-level cost data, and new-versus-returning visitor segments would further strengthen the precision of channel and page-level recommendations.